

Andre Arcaina

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Education

Toronto Metropolitan University

Sep 2022 – Apr 2027

Bachelor of Science (Honours), Computer Science (Co-op)

Coursework: Data Structures, Algorithms, Databases, Networks, Operating Systems, OOP, Distributed Systems, Big Data

Work Experience

Environment and Climate Change Canada (ECCC)

June 2025 – Present

Software Engineer Intern

North York, ON

- Modernized 10+ mission-critical Java services to Java 17 and Jakarta EE 10, improving maintainability and supporting production systems used for national weather, maritime, and Coast Guard operations.
- Utilized Graylog to analyze Java application logs and Grafana backed by Prometheus to monitor CPU, heap usage, throughput, and data rejection metrics, helping troubleshoot production issues and improve system reliability.
- Spearheaded the design and implementation of an end-to-end test suite for a satellite data pipeline using Go, automating 25 test scenarios, improving Jenkins validation, and reducing bug-triage time.
- Refactored three C++ services by replacing deprecated exception patterns, improving code clarity and reducing technical debt.
- Rebuilt a legacy VB6 application in Python/PyQt and PostgreSQL, securely processing datasets with over 50,000 records.

Undergraduate Science Society of TMU (USSTM)

Jan 2025 – Apr 2026

Backend Engineer

Toronto, ON

- Architected a contract-first Go API across 32+ endpoints using PostgreSQL, SQLc, and OpenAPI, providing a type-safe backend for the campus item vault.
- Improved release reliability by implementing 37+ unit and integration tests, achieving over 85% code coverage and preventing regressions in core application workflows.
- Built AWS S3 image storage with presigned URLs to support secure, horizontally scalable before and after item image uploads.
- Developed asynchronous email processing using Redis-backed Async workers and AWS SES for reliable notification delivery.

DataKinetics

May 2024 – Aug 2024

Software Engineer Intern

Ottawa, ON

- Built a Spring Boot REST API and web dashboard with COBOL Copybook to JSON parsing, enabling engineers to edit Db2 records and decode EBCDIC/hex data for easier mainframe inspection.
- Created 75 Postman requests with 97 automated JavaScript assertions for a mainframe REST API, validating retrieval, pagination, invalid inputs, error handling, and response data across 3 Db2-backed tables.
- Documented the dashboard architecture and workflows end-to-end in Confluence and presented to 12 R&D stakeholders.

Projects

Fafnir, Distributed Stock Trading Platform | Go, Kubernetes, NATS, PostgreSQL, Redis

GitHub

- Architected an eight microservice Go paper trading platform and load-tested it with 5,000 concurrent simulated users, averaging 1,600+ RPS with zero failures (P95: 42ms).
- Containerized Go microservices with Docker and deployed them to Kubernetes using reusable Helm templates and Minikube.
- Developed Grafana dashboards backed by Prometheus time-series metrics and Loki logs to monitor service health.

Hyperion, Distributed Key-Value Database | Go, Raft, Docker, BadgerDB

GitHub

- Built a Dockerized three-node key-value cluster in Go using Raft and BadgerDB, with linearizable reads and writes, automatic leader election, and one-node fault tolerance.
- Developed a chaos test harness covering network partitions, leader churn, SIGKILL crashes, and concurrent multi-client writes, validating cluster recovery and data consistency under failure.

Skills

Languages: Go, C++, Java, Python, JavaScript, TypeScript, SQL, Bash

Frameworks & APIs: Spring Boot, FastAPI, Flask, Node.js, React.js, Next.js, REST, gRPC

Data & Observability: PostgreSQL, Redis, BadgerDB, NATS, Grafana, Prometheus, Loki, Graylog

Infrastructure & Tools: Docker, Kubernetes, Helm, Jenkins, GitHub Actions, GCP, AWS, Git, Linux, Make